WORKSHEET

Hypergeometric probability experiments

1 A hypergeometric experiment has a population of 25 and random variable, X. If the sample size is 5 and there are 8 successes in the population, what is the domain of X? Circle the correct option.

A $X = \{1,2,3,4,5\}$

B $X = \{1,2,3,4,5,6,7,8\}$

C $X = \{0,1,2,3\}$

D $X = \{0,1,2,3,4,5\}$

2 A hypergeometric experiment has a random variable, Z, a population of 50 and a sample size of 8. If there are 6 successes in the population, what is the domain of Z? Circle the correct option.

A $Z = \{0,1,2\}$

B $Z = \{0,1,2,3,4,5,6\}$

C $Z = \{0,1,2,3,4,5,6,7,8\}$

D $Z = \{1,2,3,4,5,6,7,8\}$

3 A hypergeometric experiment has a random variable X, a population of 90 and a sample size of 14. If there are 20 successes in the population, what is $P(X = x)$? Circle the correct option.

A $\dfrac{\binom{x}{20}\binom{14-x}{70}}{\binom{90}{14}}$

C $\dfrac{\binom{20}{x}\binom{70}{14-x}}{\binom{90}{14}}$

B $\dfrac{\binom{20}{x}\binom{70}{x-14}}{\binom{90}{14}}$

D $\dfrac{\binom{14}{x}\binom{76}{20-x}}{\binom{90}{20}}$

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4 Five cards are dealt from a well-shuffled deck of 52 cards. Random variable X is the number of hearts dealt. In this experiment:

a what is the value of N?

b what is the value of n?

c what is the value of k?

d does X have a hypergeometric distribution?

5 Five cards are randomly dealt from a standard 52 card deck. Find the probability that:

a there are four kings in the five card hand

b there are five spades

c there is 1 diamond

d there are 2 aces.

6 A manufacturer takes a sample of 21 parts from a batch of 500 to test for defects. If random variable X = the number of defective parts in the sample, find the probability that:

a there are zero defects in the sample if 5% of the population is defective

b there are five defects in the sample if 8% of the population is defective

c there are 12 defects in the sample if 1% of the population is defective

d there are 12 defects in the sample if there are 100 defective parts in the population.

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7 A bag contains 20 blue marbles and 10 white marbles. If 6 are randomly chosen without replacement, find the probability that:

a 6 white marbles are drawn

b 6 blue marbles are drawn

c 3 white marbles are drawn

d 3 blue marbles are drawn.

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Answers

1 **D**

2 **B**

3 **C**

4 **a** 52

b 5

c 13

d yes

5 **a** $\approx 1.8469 \times 10^{-5}$

b $\approx 4.9520 \times 10^{-4}$

c ≈ 0.4114

d ≈ 0.0399

6 **a** ≈ 0.3329

b ≈ 0.0159

c 0

d ≈ 0.0001

7 **a** $\approx 3.5367 \times 10^{-4}$

b ≈ 0.0653

c ≈ 0.2304

d ≈ 0.2304

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